WMC Question Writing Work Sample

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Part I
Sample Proof Round Questions

1 The Cauchy Functional Equation

Problem 1a. Prove that for $f\colon \mathbb{Q} \to \mathbb{Q}$,

$$f(x) = cx \iff f(x+y) = f(x) + f(y)$$

for any $x, y \in \mathbb{Q}$. (This was a question in the WMC Finals already, but I include it here for the sake of completeness.)

Solution 1a. We begin by noting that if $f(x) = cx$ for all $x \in \mathbb{R}$, then, trivially,

$$\begin{aligned} f(x+y) &= c(x+y) \\ &= cx + cy \\ &= f(x) + f(y) \end{aligned}$$

for all $x, y \in \mathbb{R}$. That is one direction of the implication done. Now, we just need to handle the other one.

We proceed by proving that if

$$f(x+y) = f(x) + f(y)$$

for any $x, y \in \mathbb{Q}$, then $f(x) = cx$ for any $x \in \mathbb{Q}$. We do this by first doing induction over the natural numbers and then extending our result to all integers.

Let $f(1) = c$. Also, let $P(x)$ refer to the proposition that $f(x) = cx$. Then, we notice that assuming $P(x)$,

$$\begin{aligned} f(x+1) &= f(x) + f(1) \\ &= cx + c \\ &= c(x+1), \end{aligned}$$

i.e.

$$P(x) \implies P(x+1).$$

As our base case is $P(1)$, by induction, we have $P(n)$ for all $n \in \mathbb{Z}^+$.

We proceed to handling positive rational numbers. We start with reciprocals of positive integers. Notice that for $n \in \mathbb{Z}^+$,

$$\begin{aligned}
c &= f(1) \\
&= f\left(n \cdot \frac{1}{n}\right) \\
&= f\left(\frac{1}{n} + \cdots + \frac{1}{n}\right) \ (n \text{ copies}) \\
&= f\left(\frac{1}{n}\right) + \cdots + f\left(\frac{1}{n}\right) \ (n \text{ copies}) \\
&= n \cdot f\left(\frac{1}{n}\right).
\end{aligned}$$

Hence,

$$f\left(\frac{1}{n}\right) = c \cdot \frac{1}{n}.$$

Moving on to covering all rationals, notice that for $p, q \in \mathbb{Z}^+$,

$$\begin{aligned}
f\left(\frac{p}{q}\right) &= f\left(p \cdot \frac{1}{q}\right) \\
&= f\left(\frac{1}{q} + \cdots + \frac{1}{q}\right) \ (p \text{ copies}) \\
&= f\left(\frac{1}{q}\right) + \cdots + f\left(\frac{1}{q}\right) \ (p \text{ copies}) \\
&= p \cdot f\left(\frac{1}{q}\right) \\
&= c \cdot \frac{p}{q}.
\end{aligned}$$

Hence,

$$f(x) = cx$$

for all $x \in \mathbb{Q}^+$.

Proceeding to extend our result to rational numbers of all signs, notice that

$$f(0) + f(0) = f(0+0) = f(0).$$

Hence, $f(0) = 0$. Then, notice that as $f(x) = cx$ for all $x \in \mathbb{Q}^+$.

$$\begin{aligned}
f(-x+x) &= f(-x) + f(x) \\
f(0) &= f(-x) + cx \\
0 &= f(-x) + cx \\
f(-x) &= c(-x).
\end{aligned}$$

Hence,

$$f(x) = cx$$

for all $x \in \mathbb{Q}$. We will finish the problem using this intermediate result and the idea of continuity.

Problem 1b. Hence or otherwise, prove that if $f\colon \mathbb{R} \to \mathbb{R}$ is continuous, then

$$f(x) = cx \iff f(x+y) = f(x) + f(y)$$

for any $x, y \in \mathbb{R}$. (Hint: generalize your result from earlier using some calculus.)

Solution 1b. *The key intuition is that all real numbers are limits of sequences of rational numbers.* Hence, we will try to construct a sequence that approaches x for every $x \in \mathbb{R}$. Let

$$x_n = \frac{\lfloor nx \rfloor}{n}.$$

Notice that

$$nx - 1 \leq \lfloor nx \rfloor \leq nx.$$

Hence,

$$\begin{gathered} \frac{nx-1}{n} \leq \frac{\lfloor nx \rfloor}{n} \leq \frac{nx}{n} \\ x - \frac{1}{n} \leq x_n \leq x. \end{gathered}$$

From the Squeeze Theorem, as

$$\lim_{n\to\infty} \left(x - \frac{1}{n} \right) = \lim_{n\to\infty} x = x,$$

we have

$$x = \lim_{n\to\infty} x_n.$$

Therefore, as f is continuous,

$$\begin{aligned} f(x) &= f\left(\lim_{n\to\infty} x_n \right) \\ &= \lim_{n\to\infty} f(x_n) \text{ (this allows us to use } f(x) = cx \text{ for all } x \in \mathbb{Q}) \\ &= \lim_{n\to\infty} c \cdot x_n \\ &= c \cdot \lim_{n\to\infty} x_n \\ &= cx \end{aligned}$$

for all $x \in \mathbb{R}$.

2 Inequalities Through Calculus

Problem 2a. Prove that, for any f where $f'' \geq 0$ and where the following inputs are defined,

$$f(\overline{x}) \leq \sum_{i=1}^{n} a_i f(x_i),$$

where $\{a_i\}_{i=1}^n$ is a sequence of positive real numbers and $\{x_i\}_{i=1}^n$ is a sequence of real numbers,

$$\sum_{i=1}^{n} a_i = 1,$$

and

$$\overline{x} = \sum_{i=1}^{n} a_i x_i.$$

(This inequality is known as Jensen's Inequality.)

Solution 2a. We begin by considering an individual x_i and $\overline{x}$. Notice that either $x_i > \overline{x}$, $x_i < \overline{x}$, or $x_i = \overline{x}$. Also, notice that as $f'' \geq 0$, f' is increasing. (Not monotonically, though.) *The key to this problem is to use the fact that f' is increasing alongside the Mean Value Theorem. (It can also be thought of as a consideration of the area under f', which is f.)*

In the case that $\overline{x} > x_i$, from the Mean Value Theorem, for some c such that $x_i < c < \overline{x}$,

$$\frac{f(\overline{x}) - f(x_i)}{\overline{x} - x_i} = f'(c).$$

Since f' is increasing,

$$\frac{f(\overline{x}) - f(x_i)}{\overline{x} - x_i} = f'(c) \leq f'(\overline{x}).$$

Since $\overline{x} - x_i > 0$,

$$f(\overline{x}) - f(x_i) \leq f'(\overline{x})(\overline{x} - x_i).$$

In the case that $\overline{x} < x_i$, from the Mean Value Theorem, for some c such that $\overline{x} < c < x_i$,

$$\frac{f(\overline{x}) - f(x_i)}{\overline{x} - x_i} = f'(c).$$

Since f' is increasing,

$$\frac{f(\overline{x}) - f(x_i)}{\overline{x} - x_i} = f'(c) \geq f'(\overline{x}).$$

Since $\overline{x} - x_i < 0$,

$$f(\overline{x}) - f(x_i) \leq f'(\overline{x})(\overline{x} - x_i).$$

In the case that $\overline{x} = x_i$,

$$f(\overline{x}) - f(x_i) \leq f'(\overline{x})(\overline{x} - x_i).$$

is trivially true because both the left-hand and right-hand sides of the inequality are equal to zero.

Now, multiplying both sides by a_i and summing over i from 1 to n inclusive,

$$\sum_{i=1}^{n} a_i \left(f(\overline{x}) - f(x_i)\right) \leq \sum_{i=1}^{n} a_i \left(f'(\overline{x})(\overline{x} - x_i)\right)$$

$$f(\overline{x}) - \sum_{i=1}^{n} a_i f(x_i) \leq f'(\overline{x})(\overline{x} - \overline{x}) \leq 0$$

$$f(\overline{x}) \leq \sum_{i=1}^{n} a_i f(x_i).$$

Problem 2b. Let

$$\mathcal{P}(x) = \begin{cases} \left(\frac{a_1^x + a_2^x + \cdots + a_n^x}{n}\right)^{\frac{1}{x}} & \text{if } x \neq 0, \\ (a_1 a_2 \cdots a_n)^{\frac{1}{n}} & \text{if } x = 0, \end{cases}$$

where $a_1, \ldots, a_n$ are positive real numbers. Prove that $\mathcal{P}(r) \geq \mathcal{P}(s)$ if $r > s$.

Solution 2b. Notice that we are to prove that $\mathcal{P}(x)$ is an increasing function, that is, that $\mathcal{P}'(x) \geq 0$. Also, since $\mathcal{P}$ has two definitions, we would like to prove that $\lim_{x\to 0} \mathcal{P}(x) = \mathcal{P}(0)$ so that we can show that $\mathcal{P}$ is continuous.

We begin by showing that $P'(x) \geq 0$ when $x \neq 0$. *The key to this is to first consider $\ln \mathcal{P}(x)$, to get rid of the exponent $\frac{1}{x}$, and take its derivative. Then, we manipulate our inequality into a form that can be finished off with Jensen's Inequality; we do this by doing some algebraic cleanup and then using a substitution to get rid of the exponent in a_i^x.* When $x \neq 0$,

$$\ln \mathcal{P}(x) = \frac{1}{x} \ln \left(\frac{\sum_{i=1}^{n} a_i^x}{n}\right).$$

Taking the derivative of both sides,

$$\begin{aligned} \frac{\mathcal{P}'(x)}{|\mathcal{P}(x)|} &= -\frac{1}{x^2} \ln \left(\frac{\sum_{i=1}^{n} a_i^x}{n}\right) + \frac{1}{x} \cdot \frac{\sum_{i=1}^{n} \ln(a_i) \cdot a_i^x}{|\sum_{i=1}^{n} a_i^x|} \\ &= -\frac{1}{x^2} \ln \left(\frac{\sum_{i=1}^{n} a_i^x}{n}\right) + \frac{1}{x} \cdot \frac{\sum_{i=1}^{n} \ln(a_i) \cdot a_i^x}{\sum_{i=1}^{n} a_i^x}. \end{aligned}$$

Hence,

$$\begin{aligned}
&\mathcal{P}'(x) \geq 0 \\
\iff & -\frac{1}{x^2}\ln\left(\frac{\sum_{i=1}^n a_i^x}{n}\right) + \frac{1}{x}\cdot\frac{\sum_{i=1}^n \ln(a_i)\cdot a_i^x}{\sum_{i=1}^n a_i^x} \geq 0 \\
\iff & \frac{1}{x}\cdot\frac{\sum_{i=1}^n \ln(a_i)\cdot a_i^x}{\sum_{i=1}^n a_i^x} \geq \frac{1}{x^2}\ln\left(\frac{\sum_{i=1}^n a_i^x}{n}\right) \\
\iff & x\cdot\frac{\sum_{i=1}^n \ln(a_i)\cdot a_i^x}{\sum_{i=1}^n a_i^x} \geq \ln\left(\frac{\sum_{i=1}^n a_i^x}{n}\right) \\
\iff & x\sum_{i=1}^n \ln(a_i)\cdot a_i^x \geq \left(\sum_{i=1}^n a_i^x\right)\ln\left(\frac{\sum_{i=1}^n a_i^x}{n}\right).
\end{aligned}$$

Now, let $A_i = a_i^x$. Because a_i is positive, A_i is positive too. Hence, our inequality is equivalent to

$$x\sum_{i=1}^n \ln\left(A_i^{\frac{1}{x}}\right)\cdot A_i \geq \left(\sum_{i=1}^n A_i\right)\ln\left(\frac{\sum_{i=1}^n A_i}{n}\right)$$

$$\iff \sum_{i=1}^n A_i\ln(A_i) \geq \left(\sum_{i=1}^n A_i\right)\ln\left(\frac{\sum_{i=1}^n A_i}{n}\right).$$

Let $f(x) = x\ln x$ and $\overline{a} = \frac{\sum_{i=1}^n A_i}{n}$. Then, this inequality is equivalent to

$$\sum_{i=1}^n f(A_i) \geq n\cdot f(\overline{a})$$

$$\iff \frac{1}{n}\sum_{i=1}^n f(A_i) \geq f(\overline{a}).$$

Since

$$f''(x) = \frac{d}{dx}(\ln x + 1) = \frac{1}{x} > 0,$$

f is concave up, so the inequality we were considering above is true by Jensen's Inequality. Hence,

$$\mathcal{P}'(x) \geq 0$$

for $x \neq 0$. In order to show that $\mathcal{P}(r) \geq \mathcal{P}(s)$ if $r > s$, we also have to show that $\mathcal{P}$ is continuous at $x = 0$. *The motivation for this is that we are really almost done, but need to show that our inequality is not ruined by a hole at* $x = 0$.

We now proceed to show that $\lim_{x\to 0}\mathcal{P}(x) = \mathcal{P}(0)$. As before, it is helpful to consider $\ln\mathcal{P}(x)$. For $x \neq 0$,

$$\ln\mathcal{P}(x) = \frac{1}{x}\ln\left(\frac{\sum_{i=1}^n a_i^x}{n}\right),$$

so

$$\lim_{x\to 0} \ln \mathcal{P}(x) = \lim_{x\to 0} \frac{1}{x} \ln \left(\frac{\sum_{i=1}^{n} a_i^x}{n} \right).$$

Notice that

$$\begin{aligned}
\lim_{x\to 0} \ln \left(\frac{\sum_{i=1}^{n} a_i^x}{n} \right) &= \lim_{x\to 0} \ln \left(\frac{\sum_{i=1}^{n} a_i^0}{n} \right) \\
&= \lim_{x\to 0} \ln \left(\frac{n}{n} \right) \\
&= \ln 1 \\
&= 0,
\end{aligned}$$

so we have a $\frac{0}{0}$ indeterminate form, allowing us to use l'Hopital's Rule. Hence,

$$\begin{aligned}
\lim_{x\to 0} \ln \mathcal{P}(x) &= \lim_{x\to 0} \frac{\sum_{i=1}^{n} (\ln a_i) a_i^x}{\sum_{i=1}^{n} a_i^x} \\
&= \lim_{x\to 0} \frac{\sum_{i=1}^{n} \ln a_i}{n} \\
&= \lim_{x\to 0} \ln (a_1 a_2 \cdots a_n)^{\frac{1}{n}} \\
&= \ln (a_1 a_2 \cdots a_n)^{\frac{1}{n}} \\
&= \ln \mathcal{P}(0).
\end{aligned}$$

Hence,

$$\lim_{x\to 0} \mathcal{P}(x) = \mathcal{P}(0).$$

Finally, for $x \neq 0$, $\mathcal{P}$ is differentiable and hence continuous; at $x = 0$ it is continuous by the limit above. So $\mathcal{P}$ is continuous on $\mathbb{R}$, with $\mathcal{P}' \geq 0$ on $\mathbb{R} \setminus \{0\}$.

Let $r > s$. If $0 \notin (s, r)$, then $\mathcal{P}$ is continuous on $[s, r]$ and differentiable on (s, r), so by the Mean Value Theorem there is $c \in (s, r)$ with

$$\mathcal{P}(r) - \mathcal{P}(s) = \mathcal{P}'(c)(r - s) \geq 0.$$

If $s < 0 < r$, apply the same argument on $[s, 0]$ and on $[0, r]$ to get $\mathcal{P}(s) \leq \mathcal{P}(0) \leq \mathcal{P}(r)$. Either way, $\mathcal{P}(r) \geq \mathcal{P}(s)$.

3 Euler's Totient Formula Through Probability

Problem 3. Note that every $n \in \mathbb{Z}^+$ other than 1 has a unique prime factorization, which can be written as

$$n = \prod_{i=1}^{M} p_i^{k_i}$$

for some sequence of primes $\{p_i\}_{i=1}^M$ and some sequence of positive integers $\{k_i\}_{i=1}^M$. Let $\phi(n)$ be the number of integers I such that $1 \leq I \leq n$ and $\gcd(I, n) = 1$. Find a closed-form formula for $\phi(n)$.

Solution 3. *We solve this problem by considering the probability that some randomly chosen integer I such that $1 \leq I \leq n$ satisfies $\gcd(I, n) = 1$ in order to bypass the classical proof. We break $\gcd(I, n) = 1$ down into cases by considering the idea that the probabilities that any of the prime divisors p_i of n divide I are independent from each other.*

Before that, we will prove a lemma: if $a \mid n$, then $P(a \mid I) = \frac{1}{a}$ for some randomly chosen integer I such that $1 \leq I \leq n$. Let $n = ab$ for some $b \in \mathbb{Z}^+$. Then,

$$\begin{aligned} P(a \mid I) &= \frac{\#(a \mid I)}{n} \\ &= \frac{\#(I = a, I = 2a, I = 3a, \ldots, I = ab = n)}{n} \\ &= \frac{b}{n} \\ &= \frac{1}{a}. \end{aligned}$$

Therefore,

$$\begin{aligned} P\Big(\gcd(I, n) = 1\Big) &= P\Big((p_1 \nmid I) \cap (p_2 \nmid I) \cap \cdots \cap (p_M \nmid I)\Big) \\ &= P(p_1 \nmid I) \cdot P(p_2 \nmid I) \cdots P(p_M \nmid I) \\ &\quad \text{(independent by the Chinese Remainder Theorem)} \\ &= \left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_M}\right). \end{aligned}$$

Hence, as

$$\frac{\phi(n)}{n} = P\Big(\gcd(I, n) = 1\Big) = \left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_M}\right),$$

we have

$$\phi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_M}\right).$$

4 Sizes of Infinity: $|\mathbb{Z}^+| = |\mathbb{Q}^+|$

Remark. In this problem, we determine which infinite sets have the same cardinality, or size, as each other. To show that $|A| = |B|$ – that set A has the same size as set B – we can show that every $a \in A$ maps to a unique $b \in B$, and every $b \in B$ maps to a unique $a \in A$. We can specify the mapping by writing functions with the domain A and codomain B. Indeed, functions are *defined* as mappings between sets.

Problem 4a. Write a function s that establishes a bijection between $\mathbb{Z}^+$ (positive integers) and $\mathbb{Z}$ (integers), i.e. show that $|\mathbb{Z}^+| = |\mathbb{Z}|$.

Solution 4a. Let $s(n) = (-1)^n \lfloor \frac{n}{2} \rfloor$. Notice that s maps every $n \in \mathbb{Z}^+$ to a unique $m \in \mathbb{Z}$. This is because there are two n with the same $\lfloor \frac{n}{2} \rfloor$, but the parity of n determines the sign of m. Similarly, solving the equation $s(n) = m$ where $m \in \mathbb{Z}$ yields a unique $n \in \mathbb{Z}$. This is because for $m \neq 0$, there are two n such that $\lfloor \frac{n}{2} \rfloor = |m|$, and the sign of m determines the parity of n. If $m = 0$, then $n = 1$. ($n = 1$ only happens for $m = 0$, too.) Hence, *s is a bijection between $\mathbb{Z}^+$ and $\mathbb{Z}$.*

Problem 4b. Hence or otherwise, show that there exists a bijection between the positive integers and the positive rationals, i.e. that $|\mathbb{Z}^+| = |\mathbb{Q}^+|$.

Solution 4b. *The key idea is to note that both positive integers and positive rational numbers have unique prime factorizations. (We can see that positive rational numbers have unique prime factorizations by writing $q \in \mathbb{Q}^+$ in lowest terms as $\frac{a}{b}$, and writing both a and b as their prime factorizations.) We can use our function to establish a bijection between them by applying it to the exponents of the primes in the factorizations of the positive integers.* Notice that every $n \in \mathbb{Z}^+$ can be written as

$$n = p_1^{k_1} p_2^{k_2} \cdots$$

where $p_1, p_2, \ldots$ are the prime factors of n and $k_1, k_2 \cdots \in \mathbb{Z}^+$. Therefore, let $t(k) = s(k+1)$ and let

$$f(n) = \begin{cases} 1 & \text{if } n = 1, \\ p_1^{t(k_1)} p_2^{t(k_2)} \cdots & \text{if } n \neq 1. \end{cases}$$

We cannot just have s with no index shift in the exponent because $s(1) = 0$. We see that f maps every positive rational number to a unique positive integer and every positive integer to a unique rational number because t is a bijection between positive integers and integers excluding 0.